MANISH KUMAR

cs23b1044@iiitr.ac.in | +91-8887-435095 | LinkedIn | GitHub | Portfolio | Codolio

EDUCATION

Indian Institute of Information Technology, Raichur **Aug 2023 – May 2027**
Bachelor of Technology in Computer Science and Engineering **CGPA: 7.5 / 10**
Relevant Coursework: Probability & Statistics, Linear Algebra, Operating Systems, Computer Networks, Information Security Principles, Embedded AI, Machine Learning & Pattern Recognition, Data Structures & Algorithms

TECHNICAL SKILLS

- **Security & Threat Intelligence:** Threat Analysis, Anomaly Detection, Behavioral Biometrics, Attack Vector Modeling, Signature/Pattern Detection, Vulnerability Assessment, Process Sandboxing & Isolation, Incident Response Automation
- **Machine Learning & Statistical Modeling:** LSTM, CNN, Random Forest, SVM, Isolation Forest, Feature Engineering, Anomaly Scoring, Statistical Hypothesis Testing, Precision/Recall/ROC-AUC, scikit-learn, TensorFlow, Keras
- **Languages & Systems:** Python, C++, C, JavaScript, SQL, Bash/Linux, Multithreading, Memory Management, Process Control
- **Security Engineering & Tools:** REST APIs, JWT, Role-Based Access Control (RBAC), Input Sanitization, Cryptographic Hashing, Git, Docker, Postman, OpenCV, NumPy, pandas

SECURITY & MACHINE LEARNING PROJECTS

KeyProctor – AI-Powered Anomaly Detection & Sandboxed Code Execution Platform Jan 2026 – May 2026
Deep Learning (LSTM), Anomaly Detection, Keystroke Dynamics, C++ Sandboxing, Python, REST APIs Live Demo | GitHub

- Researched and developed an **AI behavioral biometrics system** analyzing keystroke dynamics (dwell time, flight time, 15+ temporal features) to detect identity impersonation and credential compromise.
- Trained an **LSTM anomaly detection model** achieving **96% accuracy** and **0.94 F1-score** (5-fold CV); deployed continuous session-level risk scoring that **reduced fraudulent access by 40%**.
- Engineered a secure, isolated **C++ sandboxed execution engine** enforcing strict process boundaries, time/memory quotas, and automated exploit/infinite loop defense for untrusted code execution.
- Implemented defense-in-depth API architecture with JWT authentication, role-based access control (RBAC), and centralized input sanitization to eliminate injection vulnerabilities.

EYeOn – Real-Time Visual Threat Detection, Surveillance & Automated Alerting Mar 2025 – May 2025
Computer Vision, Deep Face Embeddings, OpenCV, Vector Cosine Matching, Telegram Bot API Live Demo | GitHub

- Engineered a real-time biometric threat monitoring system processing multi-camera feeds with **sub-100 ms inference latency** at **15+ FPS** using multithreaded asynchronous I/O.
- Implemented 128-dimensional deep vector embeddings with adaptive cosine confidence thresholds, **cutting false-positive security alerts by 40%** across varying lighting and occlusion conditions.
- Automated incident response workflows via the Telegram Bot API, delivering actionable intrusion reports to operators within **3 seconds** of unauthorized perimeter breach detection.

TheDoctor – Secure Healthcare Platform with Threat Telemetry & Risk Modeling Jun 2025 – Aug 2025
Python, scikit-learn, Redis Caching, Threat Telemetry, MongoDB, JWT, CORS/Rate Limiting Live Demo | GitHub

- Implemented API security controls including rate limiting, CORS policies, and token encryption to safeguard sensitive health data against automated scraping and unauthorized extraction.
- Deployed an ML risk-classification pipeline (Random Forest, SVM) reaching **87% accuracy**; instrumented structured logging and error telemetry to track anomalous access patterns and maintain high availability.
- Optimized secure data retrieval pipelines with a Redis caching layer (cache-aside) and compound database indexing, reducing query response latency by **38%** (800 ms → 500 ms).

EXPERIENCE

Teaching Assistant – Data Structures & Object-Oriented Programming **Sep 2025 – Dec 2025**
Indian Institute of Information Technology, Raichur

- Mentored **120+ students** in C/C++ memory safety, pointer mechanics, buffer manipulation, OOP paradigms, and systematic debugging across bi-weekly instructional labs.
- Designed **automated vulnerability and test suites** validating edge-case inputs and boundary conditions for student submissions, accelerating evaluation turnaround by **50%** and boosting pass rates by **30%**.

ACHIEVEMENTS & LEADERSHIP

- **Algorithmic Problem Solving:** Solved **800+ problems** across LeetCode & GeeksforGeeks covering complex Graph Algorithms, Dynamic Programming, and Bit Manipulation (Codolio verified).
- **Academic Honors:** Awarded **NTSE Scholar** (National Talent Search Examination); qualified Indian National Mathematics Olympiad (PRMO/RMO level) and National Science Olympiad (top 1% nationally).
- **Leadership & Athletics:** Represented IIIT Raichur at the national Inter-IIIT Sports Tournament (16+ national institutes); organized collaborative peer problem-solving clinics.